Timur Umurzakov

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Robotics engineer with a PhD specializing in humanoid robotics, actuator design, and neural-network-based control systems. Experienced in both academic research and industrial project management. Proven ability to design complex robotic systems, develop control algorithms, and lead multidisciplinary teams. Focused on humanoid robotics, intelligent actuators, and simulation-driven robotic system design.

EDUCATION

PhD in Robotics Engineering, 2020-2026

Nazarbayev University, School of Engineering and Digital Sciences, Robotics Engineering, Astana, Kazakhstan

Thesis: "Neural-Augmented Modeling and Control of Hybrid Series-Parallel Elastic Actuators for Humanoid Robotics" (accepted without revisions, rare distinction).

- Designed hybrid SEA–PEA actuator combining BLDC motor, gearbox, and pneumatic compliance
- Developed ANN-based control framework (ESN, LSTM) for nonlinear dynamic systems
- Performed simulation-to-real validation on a physical humanoid robot

MA in Public Administration, 1997-1999

Kazakh Institute of Management, Economics and Strategic Research (KIMEP University), Almaty, Kazakhstan

Focus: Public Policy and Public Finance

BSc in Robotics Engineering, 1993-1995

Kazakh National and Technical University (KazNTU), Almaty, Kazakhstan

Underwater Robotics (transferred to KazNTU), 1987-1992

Bauman Moscow State Technical University, Moscow, Russia

EMPLOYMENT HISTORY

Research Assistant, 2021-2023

Nazarbayev University, Astana

- Conducted research in humanoid robotics and actuator design
- Designed simulation environments and control algorithms
- Designed and integrated control systems for humanoid robotic platforms
- Performed experimental validation on robotic prototypes

Teaching Assistant, 2020-2021

Nazarbayev University, Astana

- Supported instruction in robotics and engineering courses
- Assisted in laboratory sessions and student mentoring

Project Business Services Manager, 2010-2017

Kazakh Institute of Oil & Gas, Astana

- Managed large-scale engineering and infrastructure projects in the oil & gas sector, including planning, budgeting, and execution
- Ensured delivery of projects within scope, schedule, and budget constraints
- Led cross-functional teams of engineers and stakeholders across multiple projects

Project Manager, 2011-2014

KPJV Limited, Farnborough, UK

- Led international engineering projects in collaboration with global partners
- Managed project timelines, deliverables, and risk

Deputy Chairman (formerly Advisor), 2006-2009

National Innovation Fund, Almaty

- Provided strategic and analytical support on innovation initiatives
- Contributed to development of national-level technology programs

TECHNICAL SKILLS

- Robotics Simulation & Digital Twins: NVIDIA Isaac Sim / Isaac Lab, ROS2, CoppeliaSim (V-REP), PyBullet; expertise in digital twin simulation, real-time sensor integration, and robotic system integration
- Programming & Control Systems: Python, C/C++, Lua; implementation of real-time and high-frequency control systems for robotics applications
- Machine Learning & Neural Networks: Design and implementation of ANN-based controllers for actuators and robotic systems
- Embedded & Hardware Platforms: NVIDIA Jetson family (Nano, Xavier, Orin), ESP32, Raspberry Pi, Arduino; experience with high-performance real-time control loops
- Mechanical Design & CAD: FreeCAD, Autodesk Fusion 360; design and optimization of actuators and robotic mechanisms

SELECTED PUBLICATIONS AND CONFERENCES

- T. Umurzakov, S. Yessirkepov and M. Folgheraiter, "Neural-Augmented Hybrid Dynamic Modeling of a Series-Parallel Elastic Actuator," in IEEE Access, vol. 13, pp. 104630-104650, 2025, doi: 10.1109/ACCESS.2025.3580222
- T. Umurzakov, S. Yessirkepov and M. Folgheraiter, "Fast Prototyping and Testing of a New Full Scale Bipedal Robot," 2022 7th International Conference on Robotics and Automation Engineering (ICRAE), Singapore, 2022, pp. 215-221, doi: 10.1109/ICRAE56463.2022.10056190.
- S. Yessirkepov, T. Umurzakov, R. Shaimerdenov and M. Folgheraiter, "An Elastic Shoulder Joint for Humanoid Robotics Application," 2023 9th International Conference on Automation, Robotics and Applications (ICARA), Abu Dhabi, United Arab Emirates, 2023, p. 117-122, doi:10.1109/ICARA56516.2023.10125796.
- M. Folgheraiter, S. Yessirkepov, T. Umurzakov and R. Korabay, "NU-Biped-4 a Lightweight and Low-Power Consumption Full-Size Bipedal Robot," 2023 9th International Conference on

Automation, Robotics and Applications (ICARA), Abu Dhabi, United Arab Emirates, 2023, pp.38-43,doi:10.1109/ICARA56516.2023.10126025.

- M. Folgheraiter, T. Umurzakov and S. Yessirkepov, "A Servomotor with Adjustable Stiffness for Humanoid Robotics Application," ACTUATOR 2022; International Conference and Exhibition on New Actuator Systems and Applications, Mannheim, Germany, 2022, pp. 1-4.
- Yessirkepov, S.; Folgheraiter, M.; Abakov, A.; Umurzakov, T. Development of a Reduced-Degree-of-Freedom (DOF) Bipedal Robot with Elastic Ankles. Robotics 2024, 13, 172. https://doi.org/10.3390/robotics13120172
- S. Yessirkepov, T. Umurzakov and M. Folgheraiter, "Design and Analysis of a Parallel Elastic Shoulder Joint for Humanoid Robotics Application," in IEEE Access, vol. 13, pp. 8761-8778, 2025, doi: 10.1109/ACCESS.2025.3527873
- Folgheraiter, M.; Yessirkepov, S.; Umurzakov, T. NU-Biped-4.5: A Lightweight and Low-Prototyping-Cost Full-Size Bipedal Robot. Robotics 2024, 13, 9. https://doi.org/10.3390/robotics13010009

PROJECTS
Humanoid Robot (12 DOF Lower Body)

- Achieved stable squat motion using ANN-based control
- Designed and simulated humanoid robot in PyBullet
- Implemented ANN-based control strategies
- Integrated force sensors and IMU for balance control

Hybrid SEA–PEA Actuator

- Demonstrated improved modeling accuracy compared to classical methods
- Designed dual-mode actuator using gearbox backlash
- Developed ESN-based dynamic model
- Validated simulation-to-real performance